

09/02/2026



Date : _____

Page No.: _____

Assessment - 1

1. Cloud Computing Architecture

Cloud computing architecture defines how cloud services are built, delivered, and accessed. It is broadly divided into Front end and backend, connected through the internet.

1.1 Front end: This is what the user interacts with.

Components:

- Web browsers
- Mobile apps
- Thin clients
- APIs and SDKs

Responsibilities

- User authentication
- Request submission
- Displaying Results (UI/UX)

Example: Using a browser to access Google drive or an AWS-hosted web-app.

1.2 Backend (Cloud side)

This is where the actual cloud computing happens.

Core Components:

a) Application layer

- ↳ Hosts cloud based applications
- ↳ Example: Gmail, Netflix, Backend services

b) Service layer: Provides cloud services like

- ↳ Compute
- ↳ Storage
- ↳ Databases
- ↳ Networking



Date : _____

Page No.: _____

c) Cloud Runtime:

- ↳ Virtual Machines (VMs)
- ↳ Containers
- ↳ Runtime environments

d) Storage Layer:

- ↳ Object storage
- ↳ Block storage
- ↳ File storage

e) Infrastructure Layer:

- ↳ Physical servers
- ↳ Networking hardware
- ↳ Data Centers.

f) Management & orchestration:

- ↳ Resource allocation
- ↳ Load Balancing
- ↳ monitoring
- ↳ Scaling

g) Security Layer

- ↳ authentication and authorization
- ↳ Encryption
- ↳ Firewalls
- ↳ Intrusion detection

- 1.3 Middleware: acts as a bridge between front end and backend.
- ↳ Message queues
 - ↳ APIs
 - ↳ Load Balancers.

2. Key Characteristics of Cloud Computing

~~There are four main characteristics of cloud computing.~~

2.1 On-demand Self-service:

Users can provision resources automatically without human interaction.

Example: Launching an EC2 instance on AWS in minutes.

2.2 Broad Network Access:

Services are accessible over the internet via:

- ↳ Laptops
- ↳ Mobiles
- ↳ Tablets
- ↳ Workstations

2.3 Resource Pooling: Cloud providers use multi-tenant architecture.

- ↳ Resources are shared among users
- ↳ Dynamically allocated.

Example: Multiple companies using the same physical server without knowing it.

2.4 Rapid Elasticity & Scalability: Resources can be:

- ↳ Scaled up during high demand
- ↳ Scaled down when demand decreases

Example:

E-commerce websites during sales.



Date : _____

Page No.: _____

- 2.5 Measured Service (Pay-as-you-go) : Users pay only for what they use.
Measured by:
- ↳ CPU Hours
 - ↳ Storage used
 - ↳ Data transfer

- 2.6 High Availability & Reliability
- ↳ Fault tolerance
 - ↳ Data Replication
 - ↳ Disaster recovery
- Cloud services for 99.9% + uptime.

3. Cloud Service Models

Cloud computing is delivered through three main ~~separate~~ ^{service} models

3.1 Infrastructure as a Service (IaaS)

Provides virtualized computing resources ~~over~~ ^{over} the internet.

> What ~~are~~ the Provider Manages :

- ↳ Physical servers
- ↳ Networking
- ↳ Storage
- ↳ Virtualization

Examples:

- ↳ AWS EC2
- ↳ Google Compute Engine
- ↳ MS Azure Virtual Machines.

> What the User manages :

- ↳ OS
- ↳ Applications
- ↳ Middleware
- ↳ Runtime

Use Case:

- ↳ Full control over system
- ↳ Ideal for system administrators and advanced users.



Date : _____

Page No.: _____

3.2

Platform as a Service (PaaS)

Provides a platform to develop, run, and manage applications without worrying about infrastructure.

What the Provider Manages:

- ↳ Infrastructure

- ↳ OS

- ↳ Runtime

- ↳ Middleware

What the User Manages:

- ↳ Application code

- ↳ Data

Examples:

Google App Engine

AWS Elastic Beanstalk

MS Azure App services

Use Cases:

- ↳ Faster development

- ↳ No server management

- ↳ Ideal for developers

PTO



Date : _____

Page No. : _____

3.3 Software as a Service (SaaS)

Provides ready to use software applications over the internet.

What the Provider manages:

↳ Everything (Infrastructure to application)

What the User manages:

↳ The software only

Examples:

Gmail

Google Docs

Salesforce

Dropbox

Use case:

End Users

No installation or maintenance required.

4. AWS as a Cloud Server Provider

What is AWS?

Amazon Web Services is the world's largest and most widely adopted cloud platform, launched by Amazon in 2006.

4.1 Core AWS Features

a) Global Infrastructure

↳ Regional

↳ Availability Zones (isolated data centers)

↳ Edge Locations (CDN)

PTO

Ensures:

- ↳ Low Latency
- ↳ High availability
- ↳ Fault Tolerance.

b) Wide Range of Services

AWS offers 200+ services, including:

- ↳ Compute (EC2, Lambda)
- ↳ Storage (S3, EBS)
- ↳ Databases (RDS, DynamoDB)
- ↳ Networking (VPC, Route 53)
- ↳ AI/ML, IoT, Analytics

c) Scalability & Elasticity

- ↳ Auto scaling
- ↳ Load Balancers
- ↳ Serverless Computing

d) Security

- ↳ IAM (Identity Access ^{management} ~~management~~)
- ↳ Data encryption
- ↳ Network firewalls
- ↳ Compliance with global standards (ISO, GDPR)

e) Pay-as-you-go Pricing

- ↳ no upfront cost
- ↳ Pay only for usage
- ↳ cost optimization tools available

- f) High Reliability
- ↳ Built-in fault tolerance
 - ↳ Multi AZ Deployments
 - ↳ Disaster recovery support.

- g) Developer & Enterprise Friendly
- ↳ APIs
 - ↳ SDKs
 - ↳ CI/CD tools
 - ↳ Free tier for beginners.

4.2 Why AWS is Popular

- ↳ First-mover advantage
- ↳ Massive service ecosystem
- ↳ Trusted by startups and enterprises
- ↳ Strong community and documentation

5. Conclusion

Cloud computing architecture enables scalable, flexible, and cost-effective computing through layered design and virtualization.

Its key characteristics make it suitable for modern applications, while IaaS, PaaS, and SaaS address different user needs.

AWS, as a ~~one~~ leading cloud provider, offers a comprehensive, secure, and globally available platform that powers a large portion of today's internet.